Skills Summary

Product, Quotient, and Power Properties of Logarithms

Use this reference while completing your worksheet or when studying.

Skill 1 — Expanding a Logarithm into Terms

The three properties (same base b throughout, all arguments positive):

- **Product:** $\log_b(MN) = \log_b M + \log_b N$
- **Quotient:** $\log_b(M/N) = \log_b M - \log_b N$
- **Power:** $\log_b(M^p) = p \log_b M$

Expanding runs left to right: multiplication becomes addition, division becomes subtraction, an exponent becomes a coefficient.

Example: Expand $\ln(16x)$, writing the constant as a multiple of $\ln 2$.
Step 1. The argument is a product, so the product property splits it; the numeric factor is a power of 2, so the power property finishes it.
Step 2. $\ln(16x) = \ln 16 + \ln x$, and $16 = 2^4$, so $\ln 16 = 4 \ln 2$.
$\Rightarrow$ $\mathbf{4 \ln 2 + \ln x}$

Try it: Expand $\log_5\left(\dfrac{x}{125}\right)$ into two terms.

check: $\mathbf{\log_5 x - 3}$

Skill 2 — Condensing to a Single Logarithm

Rule — read the same three properties right to left. Deal with coefficients *first*: a number in front of a log is an exponent inside it. Then everything added goes into the numerator and everything subtracted goes into the denominator of one logarithm. Condensing is the move that makes an equation solvable, because $\log_b M = \log_b N$ forces $M = N$.

Example: Write $2\log_7 3 + \log_7 5$ as a single logarithm.
Step 1. There is a coefficient in front of the first term, so the power property must be applied before the two logs can be merged.
Step 2. $2\log_7 3 = \log_7(3^2) = \log_7 9$, and the sum then condenses by the product property to $\log_7(9 \cdot 5)$.
$\Rightarrow$ $\mathbf{\log_7 45}$

Try it: Write $\ln 24 - \ln 3$ as a single logarithm and simplify it.

check: $\mathbf{\ln 8}$

(!) Watch out: $\log_b M + \log_b N$ is $\log_b(MN)$, never $\log_b(M+N)$; and a coefficient becomes an exponent, so $2\log_b 3 = \log_b 9$, not $\log_b 6$.

Skill 3 — Evaluating and Solving with the Properties

Rule. To *evaluate*, condense to one logarithm and then ask what power of the base gives that argument. To *solve*, condense each side to a single logarithm, drop the logs, and solve the equation that remains — then check that every original argument is positive, since a value that makes one negative is extraneous.

Example: Evaluate $\log_4 8 + \log_4 2$ without a calculator.
Step 1. Neither term is a whole number on its own, but their product is a power of the base, so condense before evaluating.
Step 2. $\log_4 8 + \log_4 2 = \log_4(8 \cdot 2) = \log_4 16$, and $4^2 = 16$.
$\Rightarrow$ **2**

Try it: Solve $\log_2 x + \log_2 5 = \log_2 30$.

check: $\boldsymbol{x = 6}$